DIAGNOSIS WITHOUT EXECUTION: CAUSAL-GRAPH CASES FOR MULTI-TURN DEBUGGING AGENTS

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ABSTRACT

An agent that debugs *with a person* must elicit the evidence a fix depends on, avoid the attempts already known to fail, and ground its conclusion in what the user actually reported. Existing multi-turn benchmarks measure none of this directly: they either require a reproducible execution environment—which excludes precisely the environment-bound problems users bring—or script the user with a prompt and hope it stays faithful. We annotate 229 resolved public support threads as causal graphs whose nodes are (system-state, information-state) pairs and whose edges are clarifications, solutions, and the attempts each thread itself falsified. The same graph constrains the simulated user, grounds an execution-free judge, and yields metrics that need no rubric. Four frontier models separate into two tiers 0.32–0.35 apart; the split survives a judge from a different model family and reproduces on judge-free counts, while the ordering *within* a tier does not survive either. Two findings are less comfortable. On staying inside the evidence the stronger pair scores 0.64–0.71 and the weaker 0.05–0.09. And when a single authored answer is swapped for a plausible alternative, the strongest model stops being able to close the case at all 72% of the time—it does not adapt, it derails. A third finding is aimed at benchmark builders rather than at models: handing our simulator the resolved thread produces detectably leaked turns and yet moves the mean grade by less than 0.022, so an aggregate score cannot certify a user simulator, and fidelity has to come from structure rather than from numbers that look reasonable. We report the audit of our own instrument in the same spirit, including two harness defects that were being scored as model behaviour and two ablations whose conclusions reversed when we re-ran them at full corpus.

1 INTRODUCTION

Debugging with a user is an elicitation problem before it is a repair problem. The information a fix depends on is usually not in the report: in our corpus 79% of the facts a resolution turns on had to be asked for or measured. An agent that proposes early proposes blind.

Benchmarks for multi-turn software agents take one of two routes, and both lose the thing we want to measure. Execution-grounded suites (Jimenez et al., 2024) need a containerised reproduction, which systematically excludes the environment-bound faults real users report—drivers, devices, OS integration, account state. Simulated-user suites (Yao et al., 2024; Kim et al., 2025) avoid that cost by scripting the user, which moves the problem rather than solving it: a simulator that has seen the resolution can hand it over, and—as we show in Section 6.8—the aggregate score does not move when it does. A suite therefore cannot tell from its own numbers whether its user is faithful.

We take a third route. Each case is a resolved public thread annotated as a causal graph: nodes are (system-state, information-state) pairs, edges are clarifications the user can answer, solutions with required elements, and attempts this thread tried and falsified. The graph is an answer key, not a transcript to replay. The same object then does three jobs: it bounds what the simulated user may say (it speaks only from its current node), it gives an execution-free judge something to score against, and it makes several metrics mechanical—read off the transcript against the graph rather than scored by a rubric.

Table 1: Where this benchmark sits. *Real threads*: cases derived from support or issue threads rather than synthesised. *No execution*: scoring needs no reproduction environment. *Structured oracle*: progress is scored against an explicit state structure rather than a final artifact or a linear checkpoint list. *Falsified attempts*: the paths the source case tried and ruled out are annotated as first-class edges.

	real threads	no execution	structured oracle	falsified attempts
SWE-bench (Jimenez et al., 2024)	✓			
CAB (Kim et al., 2025)	✓			
Dialogue SWE-Bench (King & Flanigan, 2026)	✓			
CirrusBench (Yu et al., 2026)	✓		partial	
τ/τ^2 (Yao et al., 2024; Barres et al., 2025)		✓	✓	
JFTA-Bench (Wang et al., 2026)		✓	✓	
ExCyTIn (Wu et al., 2026)	✓	✓	✓	
This work	✓	✓	✓	✓

Our contributions:

1. A DSL and a corpus: 229 released cases over 78 repositories, each machine-validated and passed by a two-stage model review, with the falsified attempts annotated as first-class structure.
2. An evaluation protocol whose central comparison does not depend on an LLM judge, and a measurement of what the judge adds where it is used.
3. Results on four frontier models, including a counterfactual intervention study that the corpus design uniquely enables.
4. Evidence that an aggregate score cannot validate a user simulator: a simulator that leaks detectably still scores within 0.022 of a clean one. This is a constraint on how any simulated-user benchmark can argue for its own fidelity, ours included.
5. An audit of the harness itself. We report the defects we found in our own instrument and what each was worth—several would have been read as model behaviour—together with two ablations of ours whose conclusions reversed when the sample grew.

2 Related Work

Table 1 places the neighbours on four axes. Each holds some of them; the combination is what is unoccupied, and the fourth column is the one nothing else has.

Execution-grounded agent benchmarks. SWE-bench (Jimenez et al., 2024) and its descendants score a patch against a test suite. This is decisive where it applies and inapplicable where reproduction is the hard part. CAB (Kim et al., 2025) builds multi-turn cases from real GitHub issues but keeps Docker execution, and its own limitations section reports the resulting selection bias toward actively-maintained modern repositories. That the cost is affordable is itself evidence: forbidding execution outright costs a commercial agent only 1.25 points of resolve rate (Lin et al., 2026), and execution-free critics predict build status at 82% (Yadavally et al., 2025; Zhou et al., 2025). We give up execution deliberately, and buy back the environment-bound faults it excludes.

Simulated-user benchmarks. The τ-family (Yao et al., 2024; Barres et al., 2025), CAB (Kim et al., 2025), and dialogue-form SWE suites (King & Flanigan, 2026; Yu et al., 2026) place an LLM in the user's seat. Fidelity is the open question, and the field's own audits are unflattering: τ^2 found its simulator violating its instructions in 22% of one domain's dialogues (Barres et al., 2025), and swapping simulators moves measured success by up to 9 points (Seshadri et al., 2026). A simulator conditioned on the resolution can leak it, and we show in Section 6.8 that such a leak is invisible in aggregate scores—so a suite cannot certify its own simulator by observing that its numbers look sensible. We constrain the simulator with the graph rather than with a prompt, and screen for leaks directly.

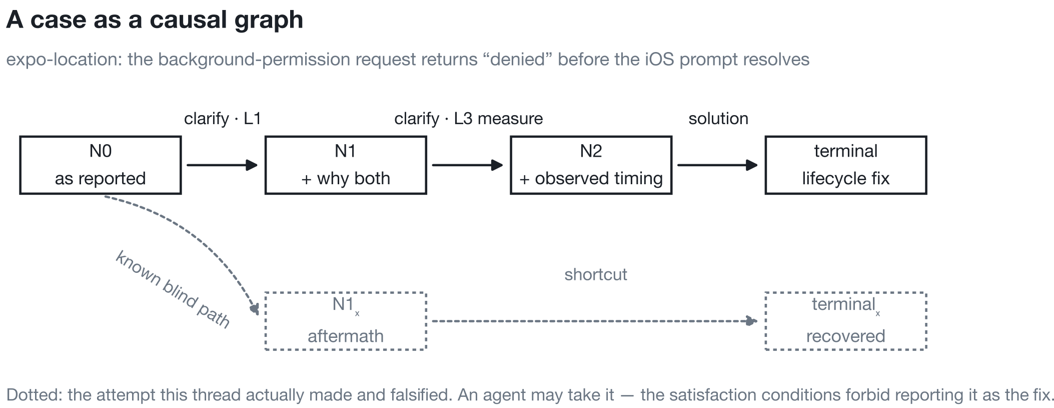

Figure 1: One case as a causal graph. Solid: the diagnostic path. Dotted: the attempt this thread actually made and falsified, together with the aftermath state it leads to. An agent may take the falsified branch—the case’s satisfaction conditions forbid *reporting* it as the fix.

Graph- and state-structured evaluation. Fault trees (Wang et al., 2026) and incident graphs (Wu et al., 2026) also score progress through structure rather than through a final artifact. Their structures are predefined domain knowledge or recorded logs; ours is extracted per-case from the thread that resolved it, and carries the attempts that thread *falsified* as first-class edges—which is what makes a counterfactual intervention well-posed.

Clarification and elicitation. A line of work scores whether a model asks a question at all, or resolves a stated ambiguity (Qian et al., 2024; Vijayvargiya et al., 2026; Qian et al., 2025; Sirdeshmukh et al., 2025); multi-turn degradation is well documented (Laban et al., 2025). We score something narrower and checkable: whether the agent obtained the specific evidence its own proposal required, at the moment it proposed.

3 Task Formulation

Figure 1 shows a complete case. A case is a directed graph over states. A state pairs a system state with an information state: what is true of the system, and what has been established in the conversation. Edges are of three kinds. *Clarification* edges carry one or more facts the user can supply, each labelled by how hard it is to obtain: **L1** any user states it unprompted, **L2** inferable from evidence already surfaced, **L3** must be asked for or measured. *Solution* edges carry an intent, a concrete example, and the elements a reply must contain to count as that fix, plus the information the fix depends on. *Known-blind* edges are solutions the thread tried and falsified; they lead to aftermath states from which the investigation continues.

Two derived layers are generated rather than authored: rollback edges out of every aftermath state, and shortcut copies of a solution edge planted at earlier states that share its system state, which let an agent skip evidence—and let the judge grade the skip.

The answer-key principle. The out-edges of a state are what *could* happen there, not a replay of what the historical participants did. This is what makes a counterfactual intervention meaningful (Section 6.5).

4 Corpus

The corpus (Figure 3, Appendix) holds 229 released cases across 78 repositories, the largest contributing 4.8%, drawn from Mozilla Bugzilla, PostgreSQL, and permissive-license GitHub repositories, threads created 2021–2026 (median 37 messages). Per-case structure and the full breakdown are in Appendix B. Cases are drafted by a model from the thread and gated by schema validators and structural lints; a two-stage review then files findings and independently verifies them. Single-pass review refuses roughly 95% of cases, mostly on claims its own evidence does not support; verification is what makes model review usable as a release gate. Of 29 cases blocked in the first round, 25 passed after redrafting under hardened rules—their defects came from the generation prompt, not from the threads.

Not recall-solvable. Handed the opening report alone, with no opportunity to ask anything, a model names both mechanism and fix class in 2.2% of cases (partial credit in a further 3.5%), with no gradient by issue year—against 28.8% for the same model allowed to converse (Figure 9). The benchmark measures interactive diagnosis, not memorisation.

5 Evaluation Protocol

The simulated user speaks only from its current graph position: that state's visible symptoms, authored answers to clarifications the agent actually asked, and—when a stall-insurance mechanism fires—a reveal that is recorded as such. It is never shown the resolution.

Execution-free describes the scoring, not the agent. What this benchmark gives up is the *reproduction environment*: there is no container to rebuild and no test suite to run, because the graph is the oracle. That is a property of how a transcript is scored, and it places no requirement on the system being scored. Whatever the agent can do—search, retrieve, open a sandbox, run an experiment of its own devising—is part of the capability under measurement, and the harness neither grants nor withholds it: an adapter receives the leak-free transcript and returns a turn, and what happens in between belongs to the agent.

What we actually ran is therefore a baseline rather than the benchmark's scope: a plain conversational model with no tools, told that the user is the one executing actions (an accurate statement of that adapter's situation, which also stops it reporting results it never obtained). Four toolless rows establish the floor and isolate elicitation from tooling. They do not characterise tool-using assistants, and we report no such claim. The release ships a second adapter that drives an external agent process—one invocation per turn, restoring its own session—so a sandboxed, retrieving or tool-using system can be measured by naming it in a command template; we provide the mechanism and no results under it.

Matching. Each agent turn is read for both acts. A question is matched against the case's clarifications; a proposal against the current state's solution edges, producing a *solution call* that records which required information the agent held at that moment, by level.

Metrics. Case outcomes (`terminal_resolved`, forced walk, ran out), turns, forced reveals per case, and premature proposals per case are read off the transcript against the graph rather than scored by a rubric. They are not model-free: the matcher deciding whether a turn asks, proposes or both, and which out-edge it hits, is itself an LLM call (Section 6.3). On top of these, four LLM rubrics—asks for evidence before proposing, stays inside the evidence, accounts for the fault, recovers from a failed step—combine with the graph-supplied grounding rate into a single `grade`.

Cost. A full row is 229 cases at a median of 20 agent turns, and the agent call dominates: median latency 105 s per turn, 117–257 agent-hours per row depending on the model. The four-row main table cost about 650 agent-hours of wall time, run six cases at a time. Token counts are not available—the gateway we measured through returns none—so we report latency, which is what a replicator will actually queue for.

Ruling on a difference. Per-case grades are paired across arms. A difference is called real only when $|t| > 2.6$ *and* a two-sided sign test gives $p < 0.01$. This replaced an earlier rule of "at least twice the run-to-run drift", which ignores sample size and consequently misruled in both directions.

6 Experiments

Table 2 gives the main result and Figure 6 its outcome composition. Two tiers separate by 0.32–0.35 ($t = 32$); within the stronger tier gpt-5.6 leads gpt-5.5 by 0.050 ($t = 5.3$), and between GLM-5.1 and Kimi-2.5 there is nothing to call. Both within-tier readings are judge-dependent and we do not claim them—see Section 6.6. Fewer than a third of cases end in a fix the agent earned, and roughly half reach a terminal state only because the insurance walked them there.

Table 2: Four models over the released corpus, 30 turns, bracketed by two scripted agents that read the graph directly (Section 6.4). The lower tier sits below the agent that asks nothing. GLM-5.1 is $n = 227$: two cases died on repeated upstream gateway timeouts and are excluded rather than scored (see Section 7).

model	grade	resolved	forced walk	ran out	turns	reveals/case
conservative (asks all)	*0.813*	210 (92%)	5 (2%)	0 (0%)	8	0.0
gpt-5.6	**0.624**	66 (29%)	110 (48%)	45 (20%)	21	3.1
gpt-5.5	**0.574**	71 (31%)	103 (45%)	47 (21%)	20	2.8
GLM-5.1	**0.306**	59 (26%)	115 (51%)	49 (22%)	22	3.7
Kimi-2.5	**0.279**	52 (23%)	103 (45%)	70 (31%)	21	4.0
blind (asks none)	*0.426*	169 (74%)	47 (21%)	6 (3%)	6	0.9

6.1 Where the weaker tier loses

Figure 4 (Appendix) localises the gap. It is not confined to any one rubric, and the largest is *staying inside the evidence*: the GPT pair scores 0.64–0.71, the other two 0.05–0.09—asserting what their conversations do not support in nearly every case.

6.2 The central comparison does not need the judge

The tier result would be weak if it existed only inside an LLM judge, so we ask whether it survives with the judge removed entirely. Two per-case quantities need no rubric to compute: forced reveals, and *premature proposals*—replies matched as a solution while the conversation sits at a state whose graph offers no solution edge, meaning the agent proposed a fix from a position where the case affords none. Both are functions of a transcript and the graph alone.

Table 3: Premature proposals per case: no judge and no rubric, though the matcher underneath is still a model (Section 6.3). The partition is the one Table 2 reports.

model	per case	comparison	difference	t	ruling
gpt-5.6	2.36	gpt-5.6 < Kimi-2.5	+1.64	6.95	real
gpt-5.5	2.22	gpt-5.6 < GLM-5.1	+1.44	6.89	real
GLM-5.1	3.79	gpt-5.5 < GLM-5.1	+1.57	7.40	real
Kimi-2.5	4.00	gpt-5.6 < gpt-5.5	−0.14	−0.82	not called
		GLM-5.1 < Kimi-2.5	+0.19	0.83	not called

Table 3 gives the result. All three cross-tier comparisons separate; neither within-tier comparison does. That is the partition the graded results produce and the partition the swapped judge produces, arrived at here by counting rather than scoring. The weaker pair proposes a fix from a state that affords none about 1.6 times more often per case, which is the behaviour the tier gap is made of.

6.3 What this removes, and what it does not

It removes the judge, not every model. Whether a turn asks or proposes, and which out-edge it hits, is decided by an LLM matcher; the counts above are mechanical given those decisions, not given the raw text. The release also carries a deterministic keyword-and-overlap matcher used as an offline fallback, and re-deriving this table with it does *not* reproduce the partition—it marks roughly four times as many proposals premature and reorders the models. We read that as a fact about the fallback, which was written as a network-free stand-in rather than a validated alternative; but it means we cannot claim this result is independent of the matcher. What we can say is that two judges from different families agree, and that both read turns segmented by the same matcher. Establishing the tier result under a competent model-free matcher is future work, and we would rather name the shared dependency than let “no judge” be read as “no model”.

6.4 What the numbers mean: a ceiling and a floor

A grade of 0.62 is uninterpretable without knowing what is achievable. We therefore run scripted agents whose behaviour is fixed by construction and read against the graph, not against a model. *Conservative* walks the canonical path and asks every clarification before proposing; *blind* never asks anything—it burns the edges this thread already falsified, then guesses down the solutions. Both are deterministic, and Table 2 brackets the model rows with them.

Two things follow. First the metrics behave as designed: the profile that asks everything grounds at 0.989 and proposes prematurely 0.004 times per case, the profile that asks nothing grounds at 0.358 and proposes prematurely 0.49 times ($t = -24.2$ and 8.3 paired). The scale is anchored, and gpt-5.6 sits about halfway up it, capturing half the headroom between asking nothing and asking everything.

Second, and less comfortably for the lower tier: GLM-5.1 and Kimi-2.5 score *below* the never-ask floor, by 0.12 and 0.15 with $|t| > 11$. Failing to elicit does not explain this—the floor also fails to elicit and does better. What differs is what they do instead: they propose from unsupported states four times per case and assert what the conversation does not carry. Confidently proposing without evidence is worse here than proposing almost nothing.

6.5 Counterfactual interventions

The corpus authors, for clarifications that mattered, an alternative answer the reporter could plausibly have given, marked with whether the right fix changes as a result. Replaying a case with one answer swapped asks whether the agent is conditioning on what the user said. Drawing 60 interventions only from cases whose baseline run closes unaided gives Figure 5 (left):

Changing one answer stops the agent closing the case at all 72% of the time (Figure 5, left). It does not switch to a different fix; it stops converging—and it derails *more* often when the change was marked as not affecting the diagnosis. On the 17 cases that stayed on the rails, five of ten interventions that should have changed the fix did.

6.6 Does the result survive a different judge?

The primary judge is from the same family as one of the models it ranks. Re-judging all four rows with GLM-5.1 under identical rubric definitions (Figure 5, right) separates two claims that a single comparison would have conflated.

The swapped judge disagrees substantially about absolute quality: it lifts the lower tier by $+0.12$ to $+0.13$ while barely moving the upper (-0.034 and -0.004), so the tier gap roughly halves. What survives is the tier structure. All three cross-tier comparisons are significant under both judges (gpt-5.6 over Kimi-2.5: $+0.345$, $t = 31.7$ primary; $+0.190$, $t = 17.8$ swapped; gpt-5.5 over GLM-5.1: $+0.269$ and $+0.134$).

The within-tier orderings are a different matter, and neither is solid enough to claim. gpt-5.6 over gpt-5.5 is significant under the primary judge ($+0.050$, $t = 5.3$) and not under the swapped one ($+0.020$, $t = 2.4$). GLM-5.1 over Kimi-2.5 clears the rule under both ($t = 2.61$ primary, $t = 3.24$ swapped)—but the primary figure passes a $|t| > 2.6$ threshold by 0.005, which is a decision-boundary artefact rather than a finding, and we decline to lean on it. We therefore claim the two tiers, which no judge in our experiments disturbs, and decline to rank within them. That a hard threshold on a continuous statistic can be crossed by five thousandths is a limitation of the ruling procedure, not evidence about the models.

6.7 What the measurement is robust to

This section previously claimed that replacing the model playing the user changes nothing, and that claim is withdrawn. It rested on -0.0007 at $n = 48$. Re-run over the full corpus the same intervention gives -0.0251 ($n = 226$, bound 0.021): swapping the simulator lowers the grade by about half as much as gpt-5.6 leads gpt-5.5, an effect this paper reports as real.

The two statistics disagree, and the disagreement is itself the finding. The paired t clears the threshold ($t = -3.08$) while the sign test does not ($p = 0.081$), so under our conjunctive rule the comparison is not called. Looking at the distribution explains why: direction is close to even (125 cases down, 98 up), but the falls are larger than the rises (-0.105 against $+0.076$). Changing the simulator does not shift every case a little; it damages a minority of cases a lot. A rule that requires consistency of direction will not register that, which is a limitation of the rule rather than evidence of no effect.

What survives is a weaker claim we still think is worth making. The literature reports simulator swaps moving measured success by up to 9 points (Seshadri et al., 2026), alongside a self-audit in which a prompted simulator violated its instructions in 22% of dialogues (Barres et al., 2025). Ours moves 2.5 points. Enumerating the user's admissible utterances in the case, rather than describing them to a prompt, appears to reduce the simulator's influence substantially—but it does not remove it, and we no longer claim that it does.

Two further interventions remain at $n = 48$: raising the turn budget from 20 to 30, and re-running the identical configuration. Both of the arms we did re-run at full corpus moved when the sample grew—this one from -0.001 to -0.025, the leak ablation from $+0.037$ to -0.011. We therefore treat the remaining $n = 48$ rows as untested rather than as nulls, and mark them as such in Figure 2.

6.8 LEAKAGE

Handing the simulator the resolved thread—the transcript-conditioned design—produces leaked turns that a lexical screen detects (3 in 916 against 0 in 913 under our production profile). Run over the full corpus, that leak has *no measurable effect on the grade*: -0.0114 ($t = -1.36$, sign $p = 0.64$, $n = 229$), with a 99% bound of 0.022.

We had run this at $n = 48$ first, where it gave $+0.037$ against a detection bound of 0.054, and expected the full corpus to resolve a suspected shortfall of power. It did not: the effect did not grow, it changed sign. The $n = 48$ reading was noise, and we report the reversal because the earlier number would otherwise stand as a near-miss that merely needed more data.

The negative result is more useful to this paper's argument than the positive one would have been. A simulator can leak the answer and still produce scores statistically indistinguishable from a clean one—the leaked turns are demonstrably there, and the aggregate moves by less than 0.022. It follows that **an aggregate score cannot be used to validate a user simulator**. A benchmark whose evidence of simulator fidelity is that its numbers look reasonable has no evidence at all, and would not notice contamination of the kind τ^2 found by direct audit in 22% of one domain's dialogues (Barres et al., 2025). Leak-safety therefore has to come from somewhere the score cannot vouch for: a structural guarantee that the simulator can only speak from its current node, plus a direct screen for when it does otherwise. That is the design this paper argues for, and this experiment is the reason the argument cannot rest on scores.

7 AUDITING THE INSTRUMENT

Figure 2 places every comparison we ran on one axis, with the same configuration run twice as the bottom row. Building this benchmark produced findings about the benchmark. We report them because several would otherwise have been read as model behaviour.

Most intervention events were our fault (Figure 7). Before repair, 44% of cases in every row triggered a forced reveal at a rate that barely varied across models—a signal belonging to the harness. Decomposed, only 7–11% reflected an agent failing a modelled path; the rest were a matcher that could not represent a turn containing both a question and a proposal, a simulator that fabricated negative results for fixes the case never modelled, and a canonical-path search that refused the falsified edges most threads pass through, killing half of every run at a median of turn 5.

Two published findings were rubric artefacts. A rubric asking whether the agent "proactively gathered info" scored every model 0.92–0.97 and supported a conclusion that elicitation was saturated. A judge-independent count contradicted it the whole time—premature proposals per case run 2.36 to 4.00 across the four models—and under a rubric that asks whether the evidence was in hand *before* the proposal, the scores track that count at Spearman -0.80. Separately, an under-specified rubric made two judges score different constructs and produced an apparent finding that the tier gap belonged to the judge; it did not survive the rubrics being written down.

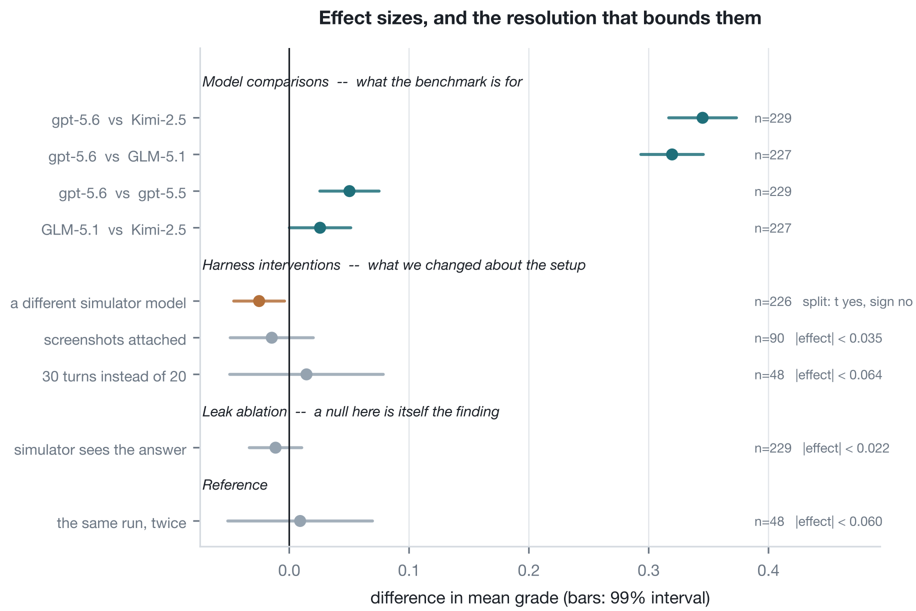

Figure 2: Every comparison, with a 99% interval, grouped by what a null would mean. Both rows we re-ran at full corpus moved: the simulator swap from -0.001 to -0.025 (orange—the paired t clears our threshold while the sign test does not, so it is not called), and the leak ablation to -0.011, bounding at 0.022 the grade effect of handing the simulator the answer, which is why the score cannot detect leakage. Rows are coloured by the paper's ruling rule rather than by whether the interval clears zero. Each uncalled row is annotated with the effect it can exclude; the remaining $n = 48$ rows exclude only ± 0.06.

A harness failure was being scored as model behaviour, in one row only. Two cases in the GLM-5.1 row produced zero-turn transcripts after the upstream gateway timed out, and the judge scored those empty transcripts 0.0 rather than skipping them—so an infrastructure failure entered the table as the model's worst possible performance, and entered only one of the four rows. The effect is small (0.303 against a clean 0.306) and its direction is what matters: it is a row-asymmetric penalty, so every comparison involving that row was biased by an amount no re-run of the other rows would reveal. We found it by auditing termination reasons per row rather than by checking that each row had 229 entries, which it did. The two cases fail reproducibly on the same upstream timeout across two further attempts, so they are excluded and the row is reported at $n = 227$ rather than repaired. The judge now has an explicit guard, and our reconciliation checks per-row termination reasons rather than row totals.

A stale summary survived the repair. Dropping those cases did not update the judgment file's cached aggregate—recomputed only when a case is judged—so it kept reporting the contaminated mean over a count it no longer held, and the repair printed that back as the new result. Our checks now recompute every aggregate from the per-case records and refuse the file when the two disagree.

Our own $n = 48$ ablations were not trustworthy, and we found out by re-running them. Two of the harness interventions in this paper were originally measured on a 48-case variance subset. Re-run over the full corpus, both moved: the leak ablation from $+0.037$ to -0.011, changing sign, and the simulator swap from -0.0007 to -0.025, turning what we had described as a clean null into an effect half the size of a model difference we report as real. Neither reversal was a change in the benchmark: the smaller sample was reporting noise inside a bound (± 0.06) loose enough to hold it. A null is a claim about a bound, and a bound three times wider than the effects under discussion supports no claim at all; the remaining $n = 48$ rows are reported as untested.

What we now check on every run. Ten offline simulator invariants; a row-integrity reconciliation of transcripts against metrics against judgments, flagging when the missing cases are disproportionately successes; a validity gate on empty replies; a leak screen; and a rubric sanity check against the behavioural counts.

8 LIMITATIONS

The judge was never validated against a human. The corpus is machine-drafted and machine-reviewed, with the audit trail published, and two checks stand in place of human validation: a judge from another family reproduces the tier ranking, and the rubrics track a judge-independent behavioural count. Neither is equivalent to correctness — two models agreeing is not the same as two models being right. What mitigates it is that the tier result is also established on judge-free counts alone, so the central comparison does not rest on the judge. The rubric-level readings do.

The instrument has a resolution, and some of our own interventions sit below it. Per-case grades are unstable across identical runs (Figure 8; mean absolute drift 0.10–0.12), so no claim here rests on an individual case. Aggregated, the 99% interval is about ± 0.025 at full corpus and ± 0.06 on the 48-case subset. The model comparisons run 5–$13\times$ the full-corpus floor and are safe. The harness interventions are not all in that position: we re-ran two of them at full corpus and both changed conclusion, so the two that remain on 48 cases are reported as untested rather than as nulls.

A tool-using agent would currently be scored unfairly, and that is a defect in the instrument rather than a gap in coverage. The grounding rubric asks how much the agent asserted as established what *this conversation* has not established. For a toolless model that is the right question, since the dialogue is its only evidence. For an agent that retrieves a release note or reads upstream source it is the wrong one: the agent is penalised for grounding a correct claim in a real source, precisely because the source is not in the dialogue. The amended construct—supported by the conversation *or* by a source the agent cites, with an uncheckable citation counted as worse than an unhedged guess—ships with the benchmark and is selectable. We report no results under it, because scores from two constructs are not comparable and this paper has already had to retract one finding that was that mistake made across two judges; the active profile is recorded in every output file for the same reason. Until a tool-using arm is run under it, these results describe elicitation in a toolless condition and should not be transferred to retrieval-augmented assistants—a real restriction, since retrieval is the intervention with the clearest route to the behaviour where the lower tier collapses.

Scope. `reached_terminal` saturates once cases use their full budget and is reported as harness health rather than as an agent metric. Two cases in the GLM-5.1 row and three in the simulator-swap arm fail reproducibly on an upstream gateway timeout and are excluded, so those rows are $n = 227$ and $n = 226$. Threads are English-only and drawn from open-source projects.

9 CONCLUSION AND RELEASE

Diagnosing a fault with a person is mostly the work of finding out what is true, and a benchmark for it has to score the finding-out rather than the final artifact. Annotating each resolved thread as a causal graph buys three things at once: the simulated user is constrained by structure instead of by instruction, several metrics become deterministic, and— because the graph records what *could* have happened rather than what did—a counterfactual intervention becomes well posed. What we find with it is that frontier models separate into two tiers by any of three independent routes, that the separation is driven by proposing fixes from positions the case does not support, and that conditioning on what the user said is fragile enough that changing one answer stops the strongest model converging on two cases in three.

The result we would most like other benchmark builders to take is the uncomfortable one: our leaking simulator scored within 0.022 of the clean one while leaking detectably, so a suite cannot use its own numbers as evidence that its user is faithful. We tried to hold ourselves to that standard, which is why the audit reports two harness defects that were being scored as model behaviour and two of our own ablations whose conclusions reversed when the sample grew.

Code ships under Apache-2.0 and the curation layer under CC BY 4.0. The 55 drafts review blocked ship beside the 229 released cases, with their findings, so the review layer can be audited rather than trusted.

Reproducibility Statement

The benchmark, the 229 released cases, the harness, and every transcript and judgment file behind the tables here are released together; the scripts that produce each figure read those files directly, so every number in this paper can be recomputed from the artifact. Case outcomes, turn counts, forced reveals and premature proposals are deterministic turn counts, forced reveals and premature proposals are mechanical functions of a transcript, a graph, and the matcher. Two things are not reproducible in the strict sense and we would rather name them than imply otherwise. Model replies are sampled, so a re-run does not reproduce a transcript; per-case grades drift by 0.10–0.12 in mean absolute terms across identical runs, which is why every claim here is an aggregate over the corpus and why the ruling procedure is a paired test. And the two GPT endpoints we measured expose no snapshot identifier (Section A), so those rows can be re-run but not pinned to a fixed model revision.

Ethics Statement

The corpus is built from public bug trackers and issue threads under permissive terms, and we redistribute thread content rather than re-hosting private data. Usernames are pseudonymised and the curation layer is released separately from the thread text so that the annotation can be reused without the identities. The people who wrote these threads were reporting bugs, not volunteering for a benchmark; we therefore restrict the release to sources whose licences permit redistribution, carry the original attribution, and will honour removal requests for individual threads. The cases describe software faults and their fixes and we see no dual-use concern beyond that. The benchmark measures diagnostic dialogue and should not be read as a safety evaluation: a model can score well here and still be unsafe to deploy in a support role.

Use of Large Language Models

LLMs are not an incidental writing aid in this work; they are part of the instrument, and we describe where. Each case in the corpus was *drafted* by a model from the source thread and then *reviewed* by a two-stage model process that files findings and independently verifies them, with schema validators and structural lints as hard gates. No case was authored by a human annotator, and the judge that produces the rubric scores is itself a model. This is the central limitation of the corpus and is discussed as such rather than buried: we release the 55 drafts that review rejected alongside the 229 that passed, so the review layer can be audited rather than trusted. The simulated user is likewise model-driven, constrained by the graph. In preparing the manuscript, an LLM was used for drafting and editing prose and for writing analysis code; the experimental design, the decisions about what to claim and what to withdraw, and the final text are the authors' responsibility.

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A Model access and version pinning

All four systems were reached through a single hosted OpenAI-compatible gateway, so the comparison is internally consistent: identical harness, identical prompts, identical judge. Exact versions are only partly pinnable, and we state which. Two endpoints resolve to reportable model identifiers (`kimi-k2.5`; a vendor-hosted GLM-5.1 deployment, which may differ from the stock release and which we name as "GLM-5.1" only for readability). The two GPT endpoints return an opaque deployment identifier and expose no snapshot string, so we cannot pin the precise revision behind the "gpt-5.6" and "gpt-5.5" rows. Readers should treat the model names as labels for the endpoints we measured rather than as verifiable version claims, and the released transcripts—not the model names—as the reproducible artifact.

Table 4: Per-case structure of the 229 released graphs. Counts are over the released corpus and are recomputable from the artifact.

	mean	median	max	total
states (nodes)	6.2	6	10	1416
transitions (edges)	5.4	5	9	1234
clarifications	5.5	5	18	1268
solution edges	2.5	2	7	575
known-blind edges	0.9	1	5	216

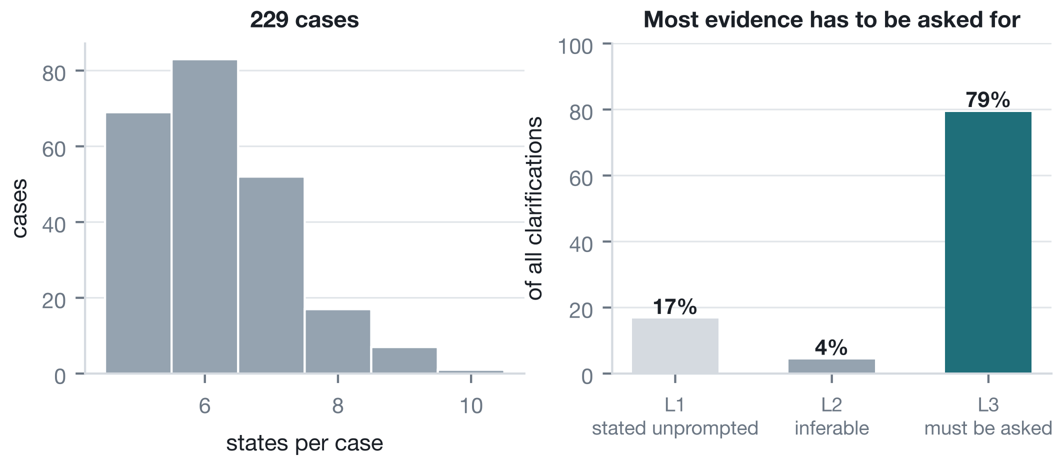

Figure 3: Corpus shape. Most evidence a resolution depends on has to be asked for: 79% of clarifications are L3.

B Corpus statistics

Edges are 659 clarification-only, 555 solution-only, and 20 mixed. Clarifications carry an obtainability level: 17% **L1** (any user volunteers it), 4% **L2** (inferable from what is already on the table), and 79% **L3** (must be asked for or measured)—the distribution behind the claim that most evidence a resolution depends on has to be elicited. 147 cases (64%) contain at least one known-blind edge, so the falsified-attempt structure is not a rare decoration: it is present in roughly two thirds of the corpus and is what makes a counterfactual intervention well posed. The 229 cases come from 78 distinct repositories with the largest contributing 11 cases (4.8%); no single project dominates.

C Supporting figures

These four figures support claims made in the main text and are placed here for space.

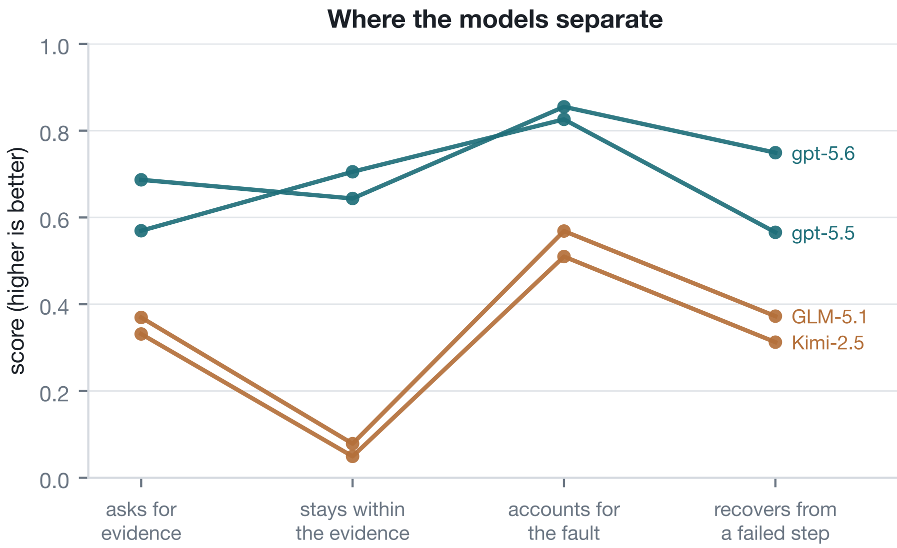

Figure 4: The tiers separate on all four rubrics and never cross. The deepest gap is staying inside the evidence.

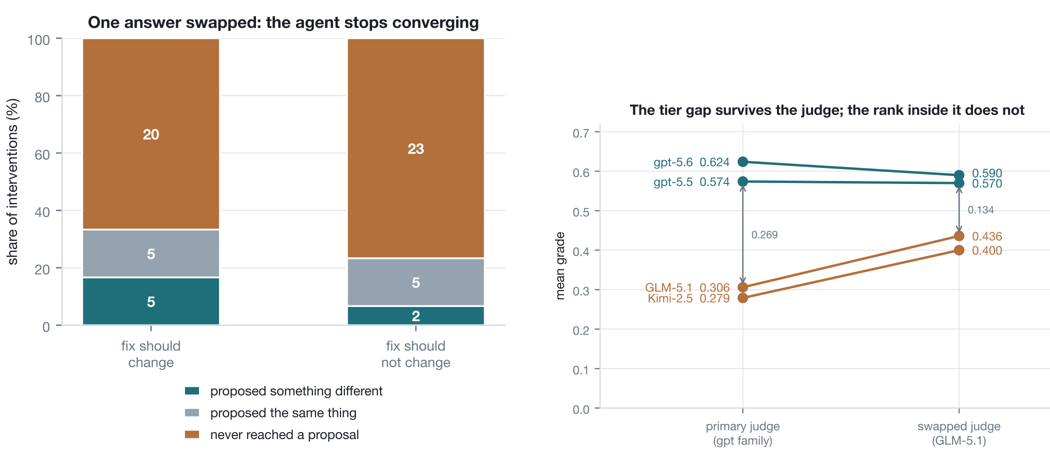

Figure 5: Left: swapping one authored answer does not redirect the agent, it stops it—and the collapse is worse in the column where the diagnosis was *not* supposed to move. Right: all four rows under two judges, same transcripts, same written rubrics. They disagree substantially about absolute quality; the tier separation survives both and the ordering within a tier does not.

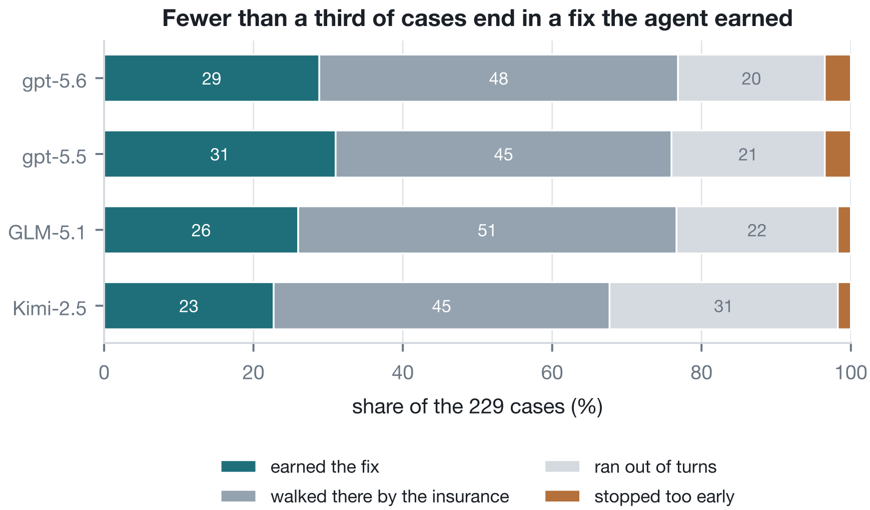

Figure 6: How cases end. The tiers differ less in fixes earned than in what happens when they fail: Kimi-2.5 exhausts its turn budget on 31% of cases against gpt-5.6's 20%, while roughly half of every row reaches a terminal state only because the stall insurance walked it there.

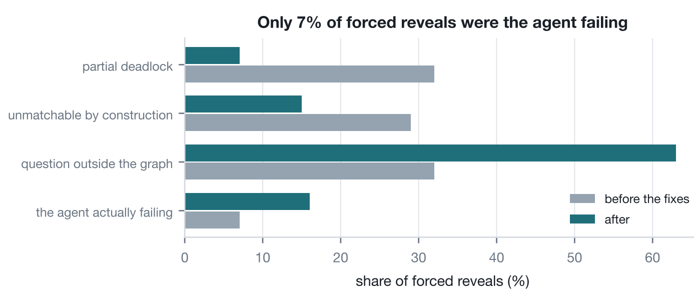

Figure 7: Decomposing the benchmark's own intervention events before repair. Read as a model metric this looked like agents stalling on half of all cases; almost all of it was the instrument.

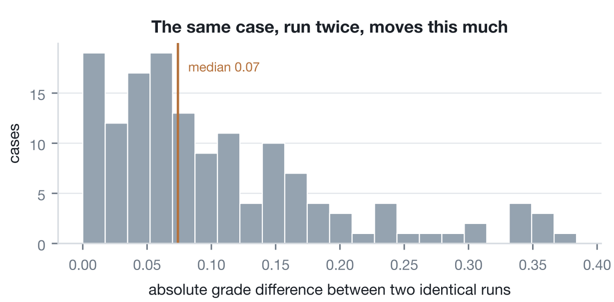

Figure 8: The same configuration run twice. Per-case grades move enough that no claim in this paper rests on an individual case; aggregate differences of the size we report survive it, and this is why the ruling procedure is a paired test rather than a threshold on the difference.

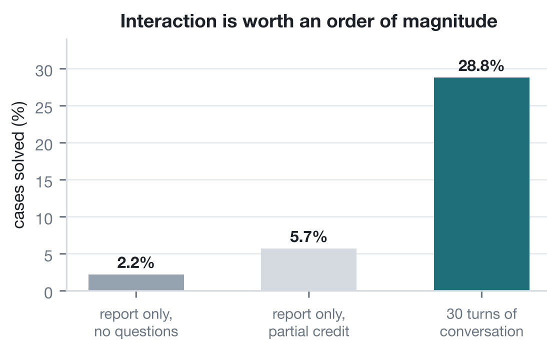

Figure 9: The same model on the same cases, with and without the ability to ask. Given the opening report alone it names mechanism and fix class in 2.2% of cases; given the conversation, 28.8%. The benchmark is measuring elicitation, not recall.